Matrix Creation 01

Given $x \in \mathbf{R}^n, y \in \mathbf{R}^p, a \in \mathbf{R}^K$, write codes to return the following:

1. $z \in \mathbf{R}^{n+p}$ such that
$$z = \begin{bmatrix} \alpha_1 x \\ \alpha_2 y \end{bmatrix}$$
where α_1, α_2 are the values that make $\|z\|_2^2 = 2\|x\|_2^2 + 5\|y\|_2^2$.
2. $U \in \mathbf{R}^{n \times K}$ such that
$$U = \begin{bmatrix} a_1 x & a_2 x & \cdots & a_k x \end{bmatrix}$$
3. $V \in \mathbf{R}^{nK \times nK}$ such that
$$V = \begin{bmatrix} a_1 x x^T & & & \\ & a_2 x x^T & & \\ & & \cdots & \\ & & & a_K x x^T \end{bmatrix}$$

You may find **repmat, broadcast_to, tile, knon (Kronecker product)** commands useful, but check if it is memory efficient.

Implementation

Goal: Fill the given function, then submit your function in Grader:

```
import numpy as np

def matcreation1(x, y, a):
    ...
    return z, U, V

if __name__ == "__main__":
    # write your test section here
    np.random.seed(42)
    x = np.random.randint(-5, 5, 3)
    y = np.random.randint(-5, 5, 4)
    a = np.random.randint(-5, 5, 2)
    z, U, V = matcreation1(x, y, a)
    print(f"z: \n {z}")
    print(f"U: \n {U}")
    print(f"V: \n {V}")
```

Input Constraints

Input: All arrays are integers $\in [-5, 5]$

- $x \in \mathbf{R}^n$ where $100 \leq n \leq 5,000$
- $y \in \mathbf{R}^p$ where $100 \leq p \leq 5,000$
- $z \in \mathbf{R}^K$ where $5 \leq K \leq 20$

Sample Output

Note: The following outputs come from the test section.

```
z: ... # REDACTED
U: [[ 1  2]
 [-2 -4]
 [ 2  4]]
V: [[ 1. -2.  2.  0.  0.  0.]
 [-2.  4. -4.  0.  0.  0.]
 [ 2. -4.  4.  0.  0.  0.]
 [ 0.  0.  0.  1. -2.  2.]
 [ 0.  0.  0. -2.  4. -4.]
 [ 0.  0.  0.  2. -4.  4.]]
```